Práctica dos: Modelo y control del sistema cardiovascular mediante el modelo Windkessel

Departamento de Ingeniería Eléctrica y Electrónica, Ingeniería Biomédica

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Información general



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Asignatura: **Modelado de Sistemas Fisiológicos**

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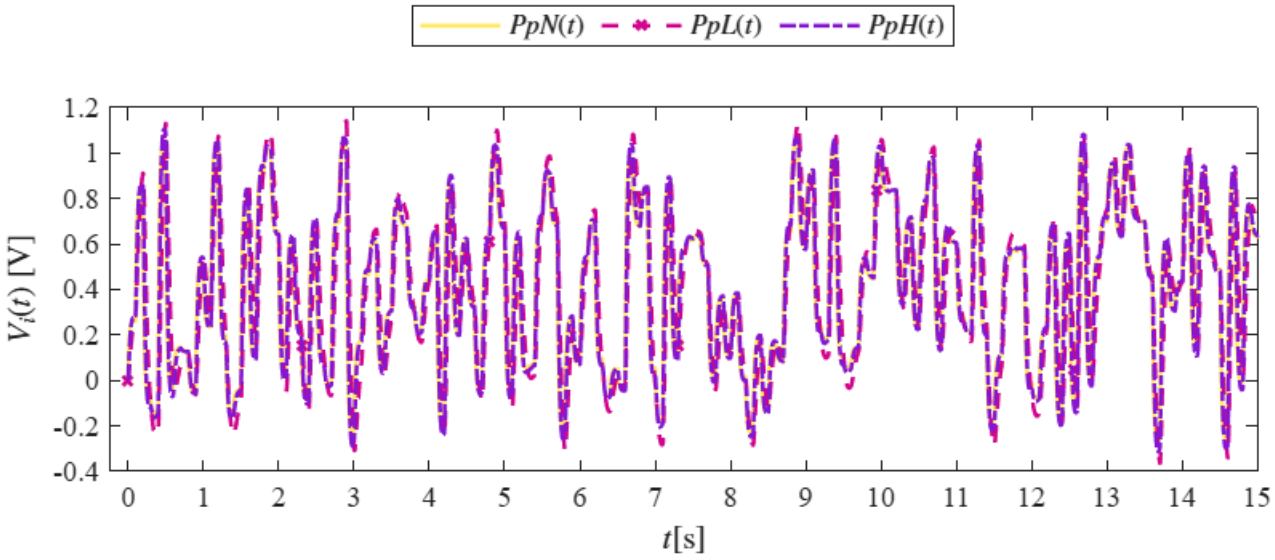
Datos de la simulación

```
clc; clear; close all; warning('off','all')
open_system('sistema3')
close_system('sistema3/Scope',0);
parameters.StopTime= '15';
parameters.Solver = 'ode15s';
parameters.MaxStep = ' 1e-2';
x = sim('sistema3',parameters);

t_data = (0:0.1:15)';
t = (0:0.001:15)';
Pa = interp1(t_data,x.Pa,t,'previous');
writematrix(Pa,'signal.xlsx');
```

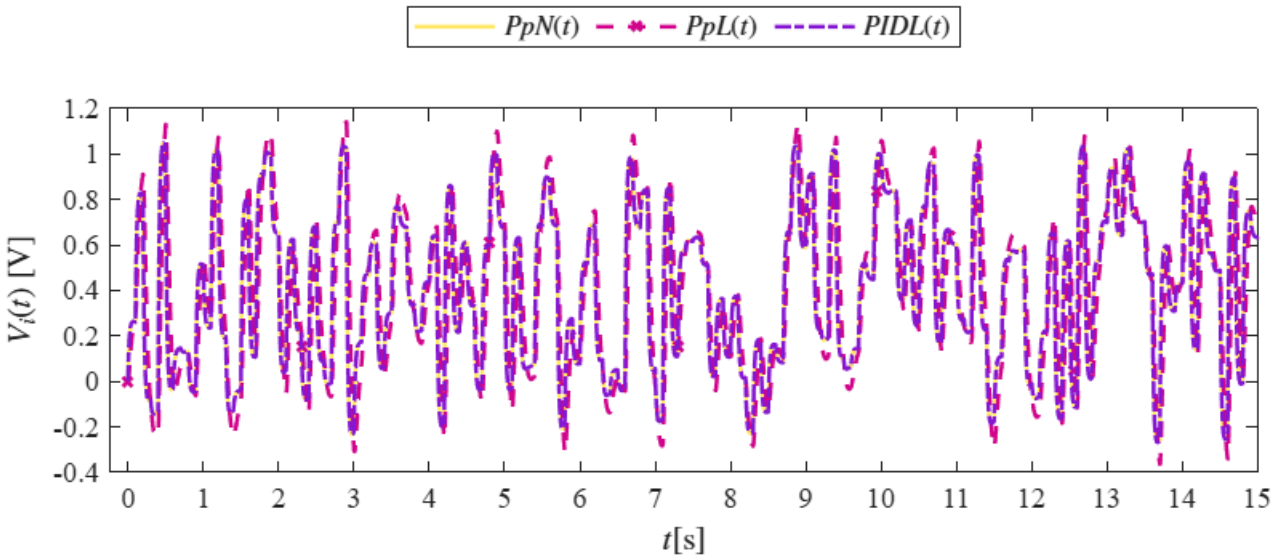
Respuesta Lazo abierto

```
plotsignals(x.t, x.PpN, x.PpL, x.PpH,x.PIDL, x.PIDH, 'Normotenso');
```



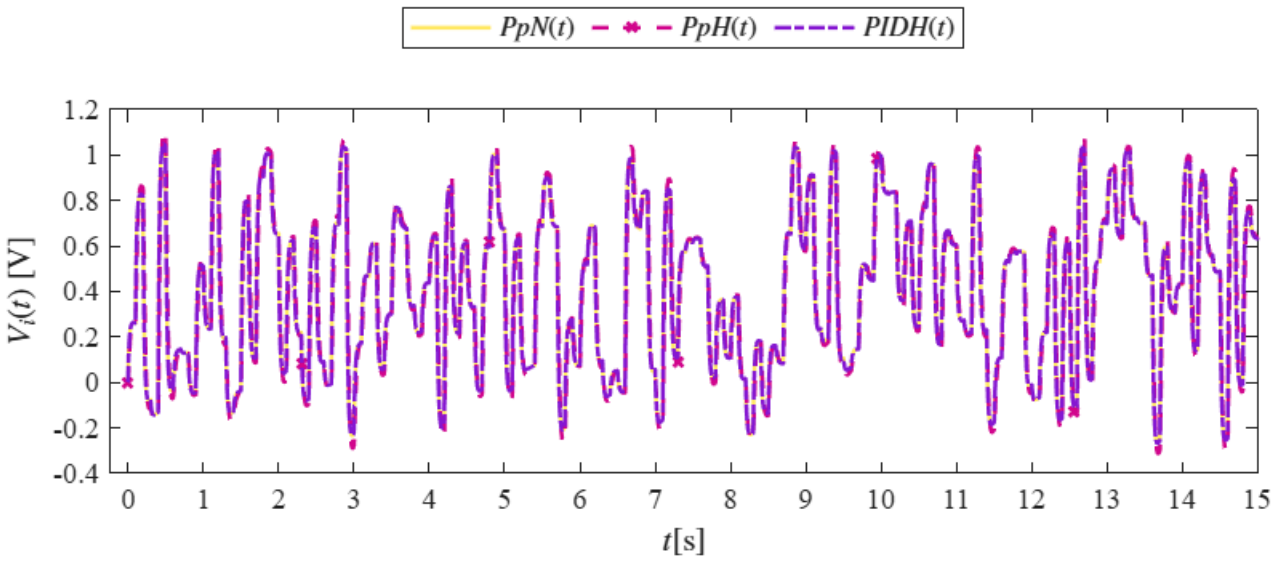
Respuesta Hipotenso

```
plotsignals(x.t, x.PpN, x.PpL, x.PpH,x.PIDL, x.PIDH, 'Hipotenso');
```



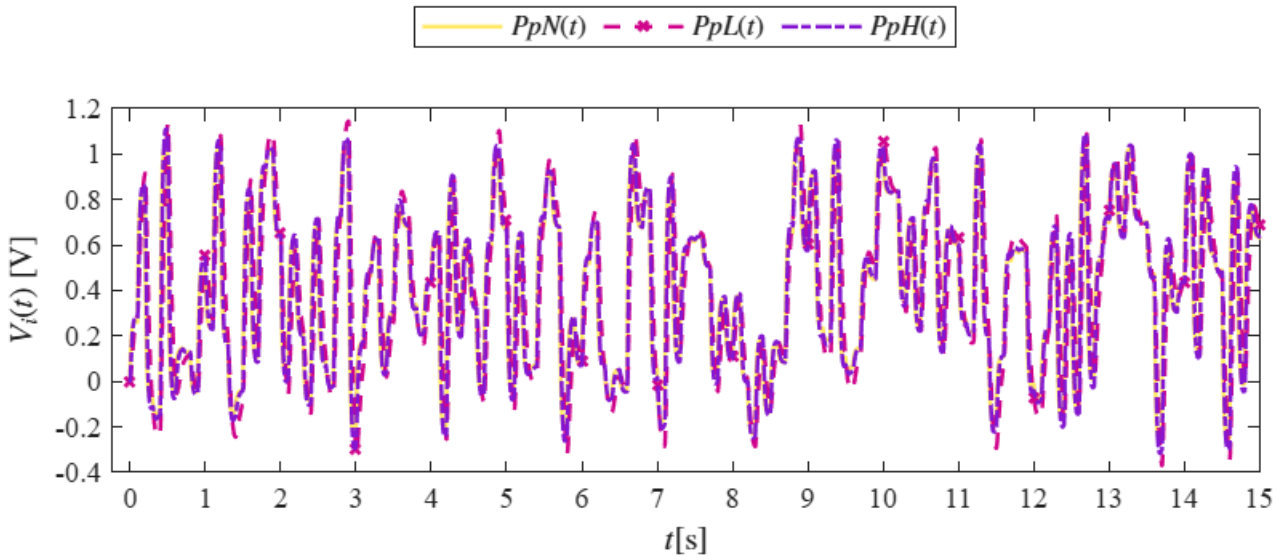
Respuesta Hipertenso

```
plotsignals(x.t, x.PpN, x.PpL, x.PpH,x.PIDL, x.PIDH, 'Hipertenso');
```



Modelo de EDOs

```
par_hipo = [0.010,0.001,1.000,3.500];
par_normo = [0.010,0.001,1.000,1.500];
par_hiper = [0.010,0.001,1.000,1.000];
PpN = cardio(par_normo,Pa,t);
PpL = cardio(par_hipo,PpN,t);
PpH = cardio(par_hiper,PpN,t);
plotsignals(t, PpN, PpL, PpH,'','','individuos');
```




```
function Pp = cardio(par,Pa,t)
Z = par(1);L = par(2);R = par(3);C = par(4);
dt = t(2);
n = round(t(end)/dt);
FL = zeros(length(t),1); FL(1) = 0;
Pp = zeros(length(t),1); Pp(1) = 0;
    for i = 1:n
        FL(i+1) = FL(i) + ((-1/L)*Pp(i) + (1/L)*Pa(i))*dt;
        Pp(i+1) = Pp(i) + ((1/C)*FL(i) - (1/(Z*C) + 1/(R*C))*Pp(i) + 1/
(Z*C)*Pa(i))*dt;
    end
end
```

Función: Respuestas de la simulación

```
function plotsignals(t, PpN, PpL, PpH, PIDL, PIDH, signal)
set(figure(), 'color', 'w')
set(gcf, 'units', 'inches', 'position', [1, 1, 25, 10])
set(gca, 'fontname', 'times new roman', 'fontsize', 11)
hold on;
grid off;
box on;
clr1 = [255, 232, 99]/255;
clr2 = [209, 5, 140]/255;
clr3 = [133, 23, 209]/255;
order = [clr1; clr2; clr3];
colororder(order)
if signal == "individuos"
    plot(t, PpN, '-', t, PpL, '--x', t, PpH, '-.', ...
        'LineWidth', 1.5, 'MarkerSize', 5, ...
        'MarkerIndices', 1:1e3:length(t));
    L = legend('$PpN(t)$', '$PpL(t)$', '$PpH(t)$');
```

```
elseif signal == "Normotenso"
    plot(t, PpN, '-', t, PpL, '--x', t, PpH, '-.', ...
        'LineWidth', 1.5, 'MarkerSize', 5, ...
        'MarkerIndices', 1:1e3:length(t));
    L = legend('$PpN(t)$', '$PpL(t)$', '$PpH(t)$');
elseif signal == "Hipotenso"
    plot(t, PpN, '-', t, PpL, '--x', t, PIDL, '-.', ...
        'LineWidth', 1.5, 'MarkerSize', 5, ...
        'MarkerIndices', 1:1e3:length(t));
    L = legend('$PpN(t)$', '$PpL(t)$', '$PIDL(t)$');
elseif signal == "Hipertenso"
    plot(t, PpN, '-', t, PpH, '--x', t, PIDH, '-.', ...
        'LineWidth', 1.5, 'MarkerSize', 5, ...
        'MarkerIndices', 1:1e3:length(t));
    L = legend('$PpN(t)$', '$PpH(t)$', '$PIDH(t)$');
end
set(L, 'interpreter', 'latex', ...
    'location', 'northoutside', ...
    'orientation', 'horizontal')
xlabel('$t$[s]', 'interpreter', 'latex')
ylabel('$V_i(t)$ [V]', 'interpreter', 'latex')

xlim([-0.25 15])
xticks(0:1:15)

ylim([-0.4 1.2])
yticks(-0.4:0.2:1.2)
exportgraphics(gcf, [signal, '_matlab.pdf'], ...
'contenttype', 'vector')
end
```